**Attachment C**

# Affidavit of Ms. Danielle Croop, Senior Manager in the Market Design group of PJM Interconnection, L.L.C.

**UNITED STATES OF AMERICA**
**BEFORE THE**
**FEDERAL ENERGY REGULATORY COMMISSION**

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| **PJM Interconnection, L.L.C.** | ) | **Docket No. ER26-____-000** |

**AFFIDAVIT OF DANIELLE CROOP**
**ON BEHALF OF PJM INTERCONNECTION, L.L.C.**

1. My name is Danielle Croop, Senior Manager in the Market Design group for PJM Interconnection, L.L.C. ("PJM"). My business address is 2750 Monroe Blvd., Audubon, Pennsylvania, 19403. I am submitting this affidavit on behalf of PJM to support the proposed reforms to PJM's Open Access Transmission Tariff ("Tariff") and PJM's Reliability Assurance Agreement Among Load Serving Entities ("RAA") to implement the Reliability Backstop Procurement ("RBP") framework by providing support for PJM's proposal to:

    a. Establish a region-wide procurement target for the RBP;
    b. Require that supply meet certain criteria to be eligible to reduce the Initial RBP Target, which differs from the criteria for supply eligible to be RBP Resources;
    c. Require that supply meet certain criteria to be eligible to submit RBP Sell Offers as RBP Resources, which differs from the criteria for supply eligible to reduce the Initial RBP Target;
    d. Rely on a pay-as-bid clearing structure based on a MW-weighted average of the levelized procurement cost cap equal to $555/megawatt ("MW")-day Unforced Capacity ("UCAP");
    e. Select resources based on a MW-weighted average of the levelized procurement cost;
    f. Not propose explicit market power screening or mitigation measures as part of the RBP;
    g. Settle RBP Resource commitments based on contract for difference ("CfD") Structure; and
    h. Proposed Cost Allocation Methodology.

    In addition, I am sponsoring one exhibit, appended to my affidavit as Exhibit 1 and entitled RBP Selection Process Examples. Exhibit 1 contains three simplified examples that help demonstrate how the RBP selection process works.

## QUALIFICATIONS

2. In my role as Senior Manager for PJM, I lead a team of Market Design Specialists and Economists on the development of PJM's market design across all of PJM's markets including the energy, ancillary services, and capacity markets. My responsibilities include leading the development of wholesale market rules and

participation frameworks that advance market efficiency and grid reliability. I joined PJM in 2014 as an Engineer, and since then, I have held numerous other roles, including as a Senior Engineer, Senior Lead Engineer, Senior Lead Market Design Specialist, and Market Design Manager, providing technical support and leading cross-divisional teams on market design, market evolution initiatives related to energy storage and distributed energy technologies participating in the PJM markets, and balancing operations at PJM, including involvement in a number of North American Electric Reliability Corporation subcommittees and initiatives.

3. Prior to joining PJM, I was employed by the SI Organization, Inc. as a System Engineer supporting large system integrations. I received a Bachelor of Engineering Degree and a Master of Engineering (Electrical Engineering) from Drexel University in Philadelphia, Pennsylvania, and have obtained a Certificate in Engineering Management.

## I. KEY DESIGN CONSIDERATIONS AND RATIONALE

4. PJM is aiming to commence a procurement of new capacity resources through the RBP[1] on September 30, 2026. PJM will set a procurement target for the RBP, calculated as the shortfall MW from the 2028/2029 Base Residual Auction ("BRA"), as adjusted to reflect voluntarily contracted new capacity procured by load coming on the PJM system in the 2027/2028 and 2028/2029 Delivery Years and any supported load that opts-out of the procurement. The shortfall MW from the 2028/2029 BRA was 6,831.3 MW of UCAP below the Regional Transmission Organization ("RTO") Reliability Requirement.

5. The RBP will function as a safety net to address the shortfall MW from the 2028/2029 BRA that has not been addressed by new load bringing their own new capacity through bilateral agreements or otherwise. The RBP is a stop-gap, one-time measure aimed specifically at addressing the resource adequacy shortfall in the PJM Region, an effort design to work in part with the load that has voluntarily brought supply to the PJM Region—independently or with support of ongoing PJM bilateral matching efforts—and a return to market fundamentals to support the expected load growth over the next decade. The RBP seeks to obtain "RBP Resources," a new, defined term in the Tariff, for a fixed, 15-year period, which will be set from the 2028/2029 Delivery Year to the 2042/2043 Delivery Year. The commitment term of the RBP Resources will vary by each resource's commercial operation date. Those RBP Resources with a commercial operation date later than 2028/2029 will have a shorter commitment, and the RBP commitments will not extend beyond the 2042/2043 Delivery Year.

6. PJM will run the procurement mechanism as a pay-as-bid, capacity-only structure, with a MW-weighted average levelized procurement cost maximum willingness to pay equal to $555/MW-day UCAP across all Delivery Years. PJM will commit

[1] All capitalized terms that are not otherwise defined herein have the meaning defined in the Tariff, Amended and Restated Operating Agreement of PJM Interconnection, L.L.C., or the RAA.

resources at an RTO-level. The total costs of the daily RBP Charges will be allocated to Load Serving Entities ("LSE") within a zone/area that had forecasted load growth, based on an RBP Obligation ("RBO") MW, which is a newly defined term that incorporates state-determined allocations of RBP UCAP MW to LSEs and end-use customers, where available, that I elaborate on further, below.

## II. REGION-WIDE PROCUREMENT OF RBP RESOURCES

7. The RBP is being driven by the resource adequacy shortfall at the RTO level. Therefore, the RBP aims to pool the supply and demand at the RTO level to address this shortfall. PJM does not propose to match specific supply to specific demand or to apply locational constraints in the central procurement.

8. PJM recognizes there are locational considerations at large for resource adequacy, and these constraints are modeled in the Reliability Pricing Model ("RPM"). RPM is an annual three-year forward framework, which allows PJM to model the necessary locational parameters—Capacity Emergency Transfer Objectives ("CETO") and Capacity Emergency Transfer Limits ("CETL")—to accurately capture the locational needs in the capacity market. By contrast, PJM does not project and calculate these parameters 15-years out, which is the eligible term of the RBP, because PJM does not have the information needed to determine such locational parameters for 15-year term capacity commitments.

9. PJM's interconnection process ensures the initial deliverability of new capacity to the RTO. Going forward, the PJM Regional Transmission Expansion Plan ("RTEP") process annually studies and addresses any transmission upgrades needed to ensure the ongoing deliverability of capacity resources. Load deliverability is also assessed as part of the RTEP process, which will identify transmission needs as load patterns change.

## III. INTENTIONALLY DIFFERENT REQUIREMENTS FOR SUPPLY ELIGIBLE TO REDUCE THE INITIAL RBP TARGET AND SUPPLY ELIGIBLE TO SUBMIT RBP SELL OFFERS AS RBP RESOURCES

10. PJM is proposing to reduce the Initial RBP Target MW by Qualifying RBP Offset UCAP MW, which is a new, defined term that represents net new capacity that meets certain qualifications. Importantly, the requirements to be Qualifying RBP Offset UCAP MW that may decrement the Initial RBP Target MW are different, and more inclusive than the requirements to qualify as RBP Resources with RBP UCAP MW that may submit sell offers in the RBP.

11. The capacity provided by Qualifying RBP Offset UCAP MW and by RBP Resources that may sell RBP UCAP MW in the RBP have different obligations and therefore, PJM is proposing different criteria for these two categories of capacity.

### A. Supply Eligible to Reduce Initial RBP Target

12. For supply to qualify as Qualifying RBP Offset UCAP MW, counterparties would be required to demonstrate the existence of commercial bilateral agreements or self-supply and show that the associated resources result in net new UCAP being made available to the PJM system. The following is a list of resources that could demonstrate net- new UCAP and would be considered in this case:

    a. New build including new resources that obtain Capacity Interconnection Rights ("CIR") transferred from deactivated resources or resources that have announced deactivation as of April 10, 2026;

    b. Uprates (incremental Installed Capacity ("ICAP")/CIR only);

    c. Repowering of deactivated generators that have retired as of April 10, 2026;

    d. Surplus interconnection (incremental only);

    e. Resources that undergo fuel-switching to another more efficient fuel type that results in a higher UCAP and CIR uprate (incremental only); and

    f. New Load Management (Emergency or Pre-emergency Demand Resources and Distributed Energy Resource ("DER")) Capacity Aggregation Resources.

13. PJM is defining this criterion broadly to capture all meaningful sources of new UCAP serving new load that contribute to resource adequacy and prevent double procuring for new load that is bringing new capacity to the system.

### B. Supply Eligible to Participate as RBP Resources in the RBP

14. Resources eligible to participate as RBP Resources in the RBP, if accepted, will be procured up through the end of a fixed, 15-year capacity commitment. PJM believes additional requirements are necessary to ensure that resources permitted to participate in the RBP align with the underlying objectives of the procurement to obtain new physical capacity for the system and demonstrate a high likelihood of successfully entering service and remaining available throughout the commitment period.

15. The RBP Resource eligibility criteria are comparatively more restrictive than the criteria for Qualifying RBP Offset UCAP MW, and specifically in addition to requiring new UCAP be brought to the system, there is a requirement that there is *new ICAP and Maximum Facility Output ("MFO")* being brought to the system, and new or transferred CIRs.[2] These additional requirements remove the eligibility of two defined sources of new UCAP that may be Qualifying New UCAP MW and

[2] CIRs can be transferred from a deactivated resource or a resource that has announced deactivation as of April 10, 2026.

eligible to reduce the Initial RBP Target. More specifically, the following are not eligible to serve as RBP Resources:

a. Surplus interconnection (incremental only); and

b. Resources that undergo fuel-switching to another more efficient fuel type that results in a higher UCAP or CIR uprate (incremental only).

16. These more restrictive eligibility requirements are intended to ensure that resources receiving long-term commitments are capable of delivering the reliability benefits that the RBP is designed to secure, reduce risk, and improve confidence in project execution.

17. The exclusion of Surplus Interconnection Service ("SIS") resources from supply eligible to serve as RBP Resources was driven by the fact that these projects do not bring any additional MFO or CIRs to the system. This is a defined requirement for resources entering through the SIS-defined Tariff provisions. Resources that do interconnect and operate under SIS do not have any rights on their own to the PJM system. By design, a surplus resource can only operate while the existing resource is in service, or no more than one year after the deactivation of the existing resource, and when the existing resource is not operating and using the interconnection service. Given the 15-year term, resources need to be able to operate independently to provide certainty of deliverability for a 15-year term. Further, these projects, while increasing UCAP in a hybrid configuration, do not have clear contributions of UCAP between the existing resource and surplus resource creating uncertainty and the need for sensitivities or estimates in both the increase system capacity and the RBP settlements.

18. This is not a complete barrier to participation of surplus resources in RBP. As mentioned above, these resources are able to participate as Qualifying New UCAP MW to reduce the Initial RBP Target. Further, these sited projects would still have an opportunity to offer into the RBP and co-locate with the existing resource and be studied by PJM for increased CIRs and MFO at the facility through the interconnection process.

19. The exclusion of resources that undergo fuel-switching to another more efficient fuel type that results in a higher UCAP and CIR uprate (incremental only) is also necessary to provide the certainty of deliverability for a 15-year term expected from an RBP Resource. Fuel switching does not bring new ICAP or new MFO on to the system. There is an anticipated UCAP increase resulting in the resource Effective Load Carrying Capability ("ELCC") class change and resulting accreditation, however the potential new UCAP captured from this resource would require some estimate of the incremental UCAP value, which may require sensitivity analyses, to determine the delivered *incremental UCAP* and RBP payment, that cannot be based on any direct increase in ICAP for the resource. This creates uncertainty in what capacity will be available to the system in the 15-year term and should be settled under the RBP.

20. Additionally, PJM has proposed more stringent requirements on Demand Resources and DER Aggregation Resources eligible to qualify as RBP Resources relative to the requirements on such resources for purposes Qualifying RBP Offset UCAP MW. Specifically, to qualify as RBP Resources, Demand Resources and DER Aggregation Resources need to have contracts for the RBP term (i.e., up to 15 years) with the identified sites that are offering as RBP Resources. This is to provide assurance that the supply will be available for the 15-year term and manage the risk of the supply being available for the full term that PJM is procuring on behalf of load in the RBP. By contrast, Qualifying RBP Offset UCAP MW does not require 15-year contracts because the commitment terms (length, price, etc.) are being managed by the commercial contract between the buyer and seller.

## IV. PAY-AS-BID STRUCTURE

21. PJM generally supports a single-clearing-price market design and continues to view uniform clearing prices as the most efficient market outcome. Uniform pricing promotes transparent price discovery, facilitates forward pricing, and supports competitive market outcomes and the PJM Region sees the benefits of that design across its existing markets.

22. However, the proposed RBP differs significantly from PJM's standard market constructs and warrants a different settlement approach which is why PJM is proposing a pay-as-bid structure for several reasons unique to this procurement.

23. First, the RBP is intended as a one-time procurement mechanism designed to address near-term resource adequacy challenges while PJM continues to focus on long-term market fundamentals and supporting continued commercial bilateral contracting. Because this procurement is not designed to be a durable or repeatable procurement, its pricing outcomes are less likely to serve as incentive market signals.

24. Second, much of the longer-term capacity that the RBP framework seeks to secure is expected to be obtained through bilateral arrangements outside of PJM's organized markets. A pay-as-bid structure more closely aligns with the economics and settlement structures of bilateral contracts and reduces the likelihood that RBP outcomes will distort or influence those private contracting activities.

25. Finally, the proposed procurement is not acquiring a uniform commodity. Although the RBP will seek a fixed, 15-year period of time for procurement and cost allocation of RBP Resources, those resources may differ based on contract duration (dependent on the projected in-service date of selected resources), commercial operation dates, and other factors such as differences in ELCC accreditation and varying UCAP across the full term of the commitment. These differences create distinct products rather than a single homogeneous offering. Establishing a uniform clearing price across such varied products would require PJM to normalize offers through potentially complex and subjective assumptions.

A pay-as-bid structure avoids those challenges while providing a straightforward and transparent procurement framework.

## V. VOLUME-WEIGHTED LEVELIZED COST PROCUREMENT MAXIMUM WILLINGNESS TO PAY OF $555/MW-DAY UCAP

26. PJM believes it is necessary to identify, for the RBP, a maximum willingness to pay because it is not reasonable to assume that load should be obligated to pay an unlimited amount for supply. As with other PJM procurement mechanisms, an upper bound helps balance reliability needs with customer cost protections.

27. In this procurement, PJM is not receiving direct buy bids or an explicit expression of loads' maximum willingness to pay, and thus PJM is proposing an administrative maximum willingness to pay that PJM believes will appropriately balance reliability needs and customer cost protections.

28. The RPM relies on the Variable Resource Requirement ("VRR") curve, which is derived from the Cost of New Entry, to establish the value PJM is willing to pay for resource commitments in a given Delivery Year. Specifically, for the 2028/2029 Delivery Year BRA, absent the temporary price collar, Point 1 on the VRR curve is $555MW-day UCAP. Because the RBP is intended to address a specific resource shortfall identified in the 2028/2029 BRA, PJM is proposing to use Point 1 of the applicable BRA VRR curve as the basis for establishing the procurement maximum willingness to pay for the RBP.

29. PJM is proposing an RBP maximum willingness to pay to be a MW-weighted average of the levelized procurement cost of all selected RBP Sell Offers to be set at $555/MW-day across all Delivery Years. This results in the cost to load being allocated for RBP Charges to not exceed $555/MW-day, on average, over the 15-year term. Using this average/budget approach to the maximum willingness to pay (in comparison to providing a price cap in each Delivery Year or a direct price cap on offers) allows more flexibility in supply sell offers, to the extent that a $555/MW-day maximum willingness to pay may not capture every resource's costs and eliminates a potential barrier to supply participation.

## VI. SELECTION PROCESS

30. PJM is proposing to calculate a levelized procurement cost for each RBP Sell Offer to determine the selection. The levelized cost of capacity is an industry standard evaluation metric used in competitive resource procurements across the country. The levelized cost calculation for evaluation addresses the fundamental issue of a simple MW-weighted average cost by accounting for the time value of money, which is critical when evaluating contracts spanning 15-year terms, as PJM is proposing in the RBP. Levelizing properly discounts future cost streams to a common basis, giving us a true economic comparison across bids. In Exhibit 1, appended to my affidavit, I provide examples, with step-by-step explanations, which illustrate how PJM's proposed selection process will work.

31. The calculation PJM will use is: Levelized Cost ($/MW-day) = Net Present Value (RBP Sell Offer price * UCAP offered)/Net Present Value (UCAP offered), with a discount rate equal to 9.5% after-tax average weighted average cost of capital ("ATWACC"). The proposed discount rate is The Brattle Group-estimate calculated value used in the 2025 Periodic Review of VRR Curve Shape and Key Parameters.[3] The purpose of discounting is not to forecast inflation or market prices, but to reflect the time value of money and the opportunity cost of invested capital. The levelized cost is intended to represent the cost per UCAP unit needed to recover investment and operating costs over the offer term, and the ATWACC represents the economically appropriate discount rate. ATWACC represents the minimum expected return required by debt and equity holders and is therefore the standard rate used to annualize long-lived infrastructure costs.

## VII. MARKET MITIGATION

32. PJM is not proposing market power screening or mitigation measures as part of the RBP. The RBP is designed as a pay-as-bid procurement in which each selected resource is compensated based on its individual offer rather than a market-clearing price established by the marginal resource. As a result, the traditional incentive and ability to exercise market power are substantially reduced.

33. While the pay-as-bid design on its own does not eliminate market power concerns, the overall proposed RBP design further mitigates market power concerns. This is driven by the one-time nature of the procurement, the voluntary participation model, the limited scope and bilateral process, and the maximum willingness to pay. In a pay-as-bid approach, unlike a uniform clearing price construct where sellers are expected to offer a marginal cost, sellers are more likely to offer the expected clearing price. Given offers will likely have less variation and information underlying offers it can be difficult and ineffective to determine offers exhorting market power.

34. The procurement is voluntary, meaning suppliers can simply choose to participate or not, and then further PJM is proposing a maximum willingness-to-pay cap, which provides a direct limit on procurement costs even in the absence of offer mitigation. The illustrative example below demonstrates the inability for one resource to impact the clearing price of any other resource in a pay-as-bid structure.

[3] *2025 Periodic Review of Variable Resource Requirement Curve Shape and Key Parameters*, PJM Interconnection, L.L.C. (Nov. 7, 2025), https://www.pjm.com/pjmfiles/directory/etariff/FercDockets/9248/20251107-er25-455-000.pdf [https://perma.cc/87GP-AYL5].

This is in contrast to a uniform clearing price market where the marginal resource will set the clearing price and all resources will receive that price. A Seller has a fleet of three resources: A, B, and C and offers accordingly:

Resource A @$10/MW-day

Resource B @$10/MW-day

Resource C @$50/MW-day

Uniform-clearing price structure:

- Resource C sets the market-clearing price at $50/MW-day,
- Resource A and Resource B also receive the market-clearing price of $50/MW-day

Pay-as-bid structure:

- Resource C receives offer price of $50/MW-day
- Resource A receives offer price of $10/MW-day
- Resource B receives offer price of $10/MW-day

35. The Request for Information ("RFI")[4] PJM conducted shows a large set of potential supply to participate in the RBP or could otherwise pursue a commercial bilateral contract (via PJM's Bilateral Matching or independently) which allows the RBP to function more like a competitive solicitation.

36. In summary, given the one-time nature of the procurement, the pay-as-bid design, the voluntary participation, and the limited scope/bilateral parallel, the implementation complexity and effectiveness would likely outweigh any corresponding benefits and therefore PJM proposes to proceed without additional market power mitigation measures, beyond the maximum willingness to pay.

## VIII. CONTRACT FOR DIFFERENCE

37. CfD is a financial settlement methodology for energy contracts designed to provide price stability to both the buyer (load) and seller (supply). It is structured around a "strike price" where the settlement or CfD is based on a difference between the market price and a committed strike price. If the strike price is *higher* than the market price, the buyer must pay the difference between the strike price and market

[4] Charles River Associates, *PJM Reliability Backstop RFI Summary*, PJM Interconnection, L.L.C. (May 27, 2026), https://www.pjm.com/-/media/DotCom/committees-groups/cifp-rbp/2026/20260527/20260527-informational-only---reliability-backstop-rfi-summary---charles-river-associates.pdf [https://perma.cc/UM8V-FAC5].

price. Alternatively, if the strike price is *lower* than the market price the seller must pay back the difference between the strike price and market price.

38. PJM is proposing to settle the RBP with a CfD-like structure, where the strike price will be the RBP offer price of the committed RBP Resource for an applicable Delivery Year, and the market price will be the weighted average RPM clearing prices for the same applicable Delivery Year.

39. PJM is proposing the CfD will only be assessed for MW that are committed in the RPM and delivered and will be capped at RBP committed MW. This is to say there will not be a CfD on RBP Resource committed MW that are not delivered in the applicable Delivery Year. This is a deviation from the purely financial CfD where the settlement would be on the committed amount, not what was physically delivered. PJM's design choice of not designing a fully financial CfD was driven by the underscoring goal of the RBP supporting reliability and resource adequacy, and intended to preserve market participation while creating a more stable capacity revenue stream for the RBP Resource.

## IX. RBP COST ALLOCATION METHODOLOGY

40. As a wholesale market operator, PJM's role is to design and administer a procurement mechanism that addresses regional reliability needs in a manner consistent with our regulatory framework. Cost allocation decisions should therefore be grounded in principles that appropriately assign costs to beneficiaries (demand) while respecting the jurisdictional boundaries between wholesale market operations and retail rate design. This jurisdictional reality was also recognized in the Statement of Principles Regarding PJM signed by the National Energy Dominance Council and the 13 governors of the PJM states, issued on January 16, 2026, where PJM's 13 governors agreed to:

> [u]se their Authorities to Allocate Costs to Data Centers and Protect Residential Customers: Use all available authorities to ensure that their state public utility commissions design rate class structures to ensure that their states' LSEs allocate their share of the cost to procure new capacity through the Reliability Backstop Auctions to new data center loads that have not otherwise procured capacity or agreed to be curtailable.[5]

41. The cost allocation structure is designed around first allocating the Final RBP Target to Electric Distributor zones/areas with forecasted load adjustment, and then further allocating the charges to LSEs based on the RBO, which permits states to determine the appropriate allocation of RBP UCAP MW to specific LSEs and/or end-use customers, thus allowing Relevant Electric Retail Regulatory Authorities ("RERRA") to appropriately assign costs to their respective retail customers, including data centers. Should the relevant RERRA(s) decline to communicate

[5] *Statement of Principles Regarding PJM*, National Energy Dominance Council, 2 (Jan. 16, 2026), https://www.energy.gov/documents/statement-principles-regarding-pjm [https://perma.cc/35HJ-W56E].

RBO MW to the relevant Electric Distributors (which would then communicate that information to PJM), PJM has proposed to apply a backup default cost allocation.

### A. Step 1: Allocating to Electric Distributor Zone/Area

42. To allocate costs, i.e., RBP Charges, PJM is proposing to determine cost responsibility by starting with the Initial RBP Target and determining how responsibility for the RBP UCAP MW in the Initial RBP Target should be allocated.

43. First, the total region-wide Initial RBP Target is allocated to each Electric Distributor zone/area in accordance with Schedule 8 of the RAA, which was revised in 2024 to incorporate Load Adjustments by Electric Distributor zone/area. More specifically, the allocation of RBP UCAP MW to specific Electric Distributor zones/areas will be based on the forecasted load growth from the 2026/2027 Delivery Year to the 2028/2029 Delivery Year and incorporate the Load Adjustments set forth in PJM Table B9b from the 2026 PJM Load Forecast[6] to ultimately determine the initial allocated pro-rata share to each relevant Electric Distributor zone/area.

44. Second, the pro-rata allocation of RBP UCAP MW to each Electric Distributor's zone/area, and the Final RBP Target, will be further adjusted to reflect reductions for Qualifying RBP Offset UCAP MW and UCAP MW representing successfully supported opt-outs from RBP in each zone/area. RBP Charges are then allocated to each Electric Distributor's zone/area based on the zone/area's pro-rata share of the total region-wide Final RBP Target.

45. An illustrative example of these adjustments and the pro-rata allocation process is detailed below.

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[6] *2026 PJM Load Forecast Report Data Tables*, PJM Interconnection, L.L.C., Tab Table B9b (Feb. 2026), https://www.pjm.com/-/media/DotCom/planning/res-adeq/load-forecast/2026-load-report-tables.xlsx [https://perma.cc/SST2-EGUZ]; *2026 PJM Load Forecast Report*, PJM Interconnection, L.L.C., 5-6 (Jan. 14, 2026), https://www.pjm.com/-/media/DotCom/library/reports-notices/load-forecast/2026-load-report.pdf [https://perma.cc/J24E-7CV7] (describing Table B9b results).

### B. Example of Allocation to Electric Distributor Zone/Area

46. The initial allocation of RBP UCAP MW to each Electric Distributor zone/area will be based on Load Adjustments, as set forth in PJM Table B9b from the 2026 PJM Load Forecast, and each zone/area's share of the Initial RBP Target of 6,831.3 MW.

| AREANAME | MW load adjustment (2028-2026) | Initial Share of RBP (%) | Initial Allocation of RBP UCAP MW |
|---|---|---|---|
| BGE | 25.0 | 0.28% | 18.9 |
| PECO | 264.0 | 2.92% | 199.6 |
| PPL | 1,085.0 | 12.01% | 820.5 |
| PSEG | 320.0 | 3.54% | 242.0 |
| AEPOHIO | 2,019.2 | 22.35% | 1,527.0 |
| PE | 154.0 | 1.70% | 116.5 |
| PP | 17.8 | 0.20% | 13.5 |
| OHIO | 149.2 | 1.65% | 112.8 |
| COMED | 1,173.0 | 12.99% | 887.1 |
| DAY | 277.0 | 3.07% | 209.5 |
| DOM | 982.6 | 10.88% | 743.1 |
| NVEC | 1,722.4 | 19.07% | 1,302.6 |
| ODEC | 170.2 | 1.88% | 128.7 |
| REC | 673.8 | 7.46% | 509.6 |
| RTO | 9,033.2 | 100% | 6,831.3 |

47. Second, as decrements are submitted, both the Final RBP Target and the pro-rata allocation to each Electric Distributor zone/area will be updated. In this example, I will assume the following Qualifying RBP Offset UCAP MW were submitted and will decrement the Initial RBP Target:

| AREANAME | Load | Qualifying RBP Offset UCAP MW |
|---|---|---|
| REC | 673.8 | 633.4 |
| ODEC | 130 | 122.2 |

Note: The qualifying RBP Offset UCAP MW is represented by load*FPR for the 2028/2029 BRA (0.9401). This is representative of the self-supply obligation.

48. In my illustrative example, for purposes of simplicity, I assume there are no additional decrements to the Initial RBP Target for opt outs. Thus, the Final RBP Target would be calculated as Initial RBP target – Qualifying RBP Offset UCAP MW. In this example, this would be: 6,831.3 MW – [633.4 MW + 122.2 MW ] = 6,075.7 MW. The Qualifying RBP Offset UCAP MW is divided by the 2028/2029 BRA Forecast Pool Requirement to identify the amount of load that is being served by those offset MW and can subsequently be removed from any allocation of RBP MW.

49. The updated allocation would be as follows, as demonstrated in the table below. In the Old Dominion Electric Cooperative ("ODEC") zone/area, the MW Load Adjustment is reduced from 170.2 MW to 40.2 MW to account for the 130 MW of load with demonstrated Qualifying RBP Offset UCAP MW. In the Rappahannock Electric Cooperative ("REC") zone/area, the MW Load Adjustment is reduced to 0 MW, from 673.8 MW to 0 MW to account for the 673.8 MW of load with demonstrated Qualifying RBP Offset UCAP MW.

50. Based on the revised adjustments to the Load Adjustments, each Electric Distributor zone/area's allocated RBP UCAP MW will reflect its adjusted pro-rata share of the 6,075.7 MW Final RBP Target. Thus, while ODEC had previously been allotted 1.88% of the Initial Target, its pro-rata share was reduced to 0.49% to reflect the fact that ODEC brought 122.2 MW of Qualifying RBP Offset UCAP MW, which is, with the Forecast Pool Requirement adjustment, equivalent to 130 MW. Similarly, REC's pro-rata share went from 7.46% to 0.0% to reflect that REC brought 633.4 MW UCAP supply, which is obligation for 673.8 MW of load, and is equal to the amount of the REC Load Adjustment.

| AREANAME | Final MW load adjustment (2028-2026) | Final Share of RBP (%) | Final Allocation of RBP UCAP MW |
|---|---|---|---|
| BGE | 25.0 | 0.30% | 18.5 |
| PECO | 264.0 | 3.21% | 194.9 |
| PPL | 1,085.0 | 13.18% | 801.0 |
| PSEG | 320.0 | 3.89% | 236.3 |
| AEPOHIO | 2,019.2 | 24.54% | 1,490.8 |
| PE | 154.0 | 1.87% | 113.7 |
| PP | 17.8 | 0.22% | 13.1 |
| OHIO | 149.2 | 1.81% | 110.2 |
| COMED | 1,173.0 | 14.25% | 866.0 |
| DAY | 277.0 | 3.37% | 204.5 |
| DOM | 982.6 | 11.94% | 725.4 |
| NVEC | 1,722.4 | 20.93% | 1,271.6 |
| ODEC | 40.2 | 0.49% | 29.7 |
| REC | 0.0 | 0.00% | 0.0 |
| RTO | 8,229.4 | 100% | 6,075.7 |

### C. Step 2: Allocating to LSEs and/or End-Use Customers Based on RERRA-Determined Allocation or Allocating to LSEs Based on PJM Backup Allocation

51. Electric Distributors will report to PJM, where available, the RBO MW for the LSEs in an Electric Distributor's zone/area that are being allocated RBP UCAP MW. These value(s) will be provided to PJM by the Electric Distributor designated to coordinate with the states and/or RERRAs to the extent that states and/or RERRAs wish to provide the Electric Distributors with information on how costs are allocated to LSEs. To the extent that PJM assigns costs to LSEs in accordance with RBO MW, PJM does so at the direction of the submitted information by relevant the Electric Distributor(s) as determined by the RERRA for that zone/area.

52. If PJM does not receive RBO MW from an Electric Distributor for a zone/area allocated RBP UCAP MW, PJM will need a backup settlement allocation. PJM will assign the RBP UCAP MW using the Daily Unforced Capacity Obligation MW load ratio shares in the respective zone/area that are assigned RBP UCAP MW.

### D. Example of Allocation to Electric Distributor Zone/Area

53. Another illustrative example is detailed below for the pro-rata allocation process to the LSEs that are being allocated RBP UCAP MW. In this example, we assume the Electric Distributor zone/area 1 has 116 RBP UCAP MW obligation with an associated $11,600/day RBP charge, allocated to that zone/area. There are two

LSEs (i.e., LSE 1 and LSE 2) in zone/area 1 that bear some responsibility for the RBP charge, based on the RBO submitted to PJM.

54. In the first scenario, the designated RBO MW that should be used to allocate charges to each LSE is submitted to PJM by the Electric Distributor and then PJM will apply the cost allocation and settle RBP Charges accordingly. Here, the reported RBO MW for LSE 1 is 58 RBP UCAP MW, for LSE 2 is 58 RBP UCAP MW, and for LSE 3 is 0 RBP UCAP MW. Thus the $11,600/day RBP Charges will be allocated pro rata 50% to LSE 1, 50% to LSE2, and 0% to LSE 3.

| ED zone/area 1 = $11,600/ Day RBP Charge associated with 116 RBP UCAP MW | | | | |
|---|---|---|---|---|
| LSE | RBO MW | Pro-rata | | RBP Charge ($) |
| LSE 1 | 58 | 50% | $ | 5,800.00 |
| LSE 2 | 58 | 50% | $ | 5,800.00 |
| LSE 3 | 0 | 0 | $ | - |

55. In the second scenario, where the submitted RBO MW is not equal to the RBP UCAP MW procured for the zone/area, PJM will still use the submitted RBO MW to determine the pro rata allocation and ensure the RBP Charges are fully allocated to the LSE's with non-zero submitted obligations. In the example below, LSE 1 is 60 RBP UCAP MW, LSE 2 is 40 RBP UCAP MW, and LSE 3 is 0 RBP UCAP MW. Thus, the $11,600/day RBP Charge will be allocated 60% to LSE1 and 40% to LSE 2, and 0% to LSE 3.

| ED zone/area 1 = $11,600/Day RBP Charge associated with 116 RBP UCAP MW | | | | |
|---|---|---|---|---|
| LSE | RBO MW | Pro-rata | | RBP Charge ($) |
| LSE 1 | 60 | 60% | $ | 6,960.00 |
| LSE 2 | 40 | 40% | $ | 4,640.00 |
| LSE 3 | 0 | 0 | $ | - |

56. This concludes my affidavit.

**Exhibit 1**

# Appendix to Affidavit of Ms. Danielle Croop

## Examples Demonstrating RBP Resource Procurement Process

## Examples Demonstrating RBP Resource Procurement Process

The following examples are intended to demonstrate how RBP Resources will be procured based on a maximum willingness to pay, defined as a "MW-weighted average of the levelized procurement cost" for the RBP, and selected through a "budget approach."[1]

Each RBP Resource Sell Offer will include an offer price and MW quantity pair for each Delivery Year beginning with the first year the resource is expected to reach commercial operation. PJM will then determine the "levelized procurement cost" for this resource to use in the selection process.[2] This will be done with the Levelized Cost equation, with a discount rate equal to 9.5% after-tax average weighted average cost of capital ("ATWACC").

Levelized Cost ($/MW-day) = Net Present Value (RBP Sell Offer price x UCAP offered)/Net Present Value (UCAP offered)

## Example 1

In the example below, Unit A offers at a price and a quantity for each Delivery Year of the 15-year fixed term, beginning in Year 1 and ending in Year 15. This Offer has a levelized procurement cost of $101.50/MW-day UCAP and a levelized MW value of 69.3 MW.

[1] Proposed Tariff, Attachment DD, section 18.5(f)(ii); Croop Aff. ¶ 29.

[2] Proposed Tariff, Attachment DD, section 18.5(f)(ii); Croop Aff. ¶ 29.

*Example 1: Unit A, $/MW Offer per Year*

| Unit A | Price ($) | MW |
|---|---|---|
| Year 1 | 91 | 10 |
| Year 2 | 91 | 10 |
| Year 3 | 100 | 9 |
| Year 4 | 100 | 9 |
| Year 5 | 100 | 9 |
| Year 6 | 100 | 9 |
| Year 7 | 100 | 9 |
| Year 8 | 110 | 8 |
| Year 9 | 110 | 8 |
| Year 10 | 110 | 8 |
| Year 11 | 110 | 8 |
| Year 12 | 110 | 8 |
| Year 13 | 110 | 8 |
| Year 14 | 110 | 8 |
| Year 15 | 110 | 8 |

## Example 2

Example 2 walks through how a set of offers are evaluated based on ranked Delivery Year and levelized cost for selection. The selection process is also constrained to the maximum willingness to pay of the procurement. For illustrative purposes, rather than a maximum willingness to pay of $555/MW-day as PJM is proposing to use for the RBP, we assume that the maximum willingness to pay is $100/MW-day, and the Final RBP Target for procurement is 6,800 MW.

Table 1 illustrates the UCAP MW offered through the first four Delivery Years and a represented levelized cost offer for this resource. Table 1, below, tracks the MW that are procured to determine when PJM satisfies the procurement target of 6,800 MW.

*Table 1*

| | 28/29 | 29/30 | 30/31 | 31/32 | Levelized Cost Offer |
|---|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 | $100 |
| **Offer 2** | 500 | 450 | 450 | 400 | $50 |
| **Offer 3** | | 400 | 500 | 500 | $200 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 | $50 |
| **Offer 5** | | 900 | 1,000 | 1,000 | $90 |
| **Offer 6** | | | 150 | 150 | $1,200 |
| **Offer 7** | | | | 200 | $1,500 |
| **Offer 8** | | | | 200 | $2,000 |
| **Total Offered** | 3,500 | 6,750 | 6,800 | 7,250 | |

Table 2 shows the offers in greater detail in determining the present value and levelized cost of the offers. For each RBP Resource offer (Offers 1-8), there are three procurement values in the PV column. The first value for each RBP Resource offer is the levelized MW UCAP. The second value is the levelized cost, expressed as $/MW-day. And the third value is the total revenue cost stream based on the RBP Resource's levelized MW and cost.

*Table 2*

| | | 28/29 | 29/30 | 30/31 | 31/32 | PV |
|---|---|---|---|---|---|---|
| **Offer 1** | **MW UCAP** | 3,000 | 3,000 | 2,800 | 2,800 | 9,322 |
| | **$/MW-Day** | $100 | $100 | $100 | $100 | $100 |
| | **Revenue Cost Stream** | $300,000 | $300,000 | $280,000 | $280,000 | $932,200 |
| **Offer 2** | **MW UCAP** | 500 | 450 | 450 | 400 | 1,453 |
| | **$/MW-Day** | $50 | $50 | $50 | $50 | $50 |
| | **Revenue Cost Stream** | $25,000 | $22,500 | $22,500 | $20,000 | $72,645 |
| **Offer 3** | **MW UCAP** | - | 400 | 500 | 500 | 1,062 |
| | **$/MW-Day** | $0 | $200 | $200 | $200 | $200 |
| | **Revenue Cost Stream** | $0 | $80,000 | $100,000 | $100,000 | $212,444 |
| **Offer 4** | **MW UCAP** | - | 2,000 | 1,900 | 2,000 | 4,506 |
| | **$/MW-Day** | $0 | $50 | $50 | $50 | $50 |
| | **Revenue Cost Stream** | $0 | $100,000 | $95,000 | $100,000 | $225,316 |
| **Offer 5** | **MW UCAP** | - | 900 | 1,000 | 1,000 | 2,208 |
| | **$/MW-Day** | $0 | $90 | $90 | $90 | $90 |
| | **Revenue Cost Stream** | $0 | $81,000 | $90,000 | $90,000 | $198,705 |
| **Offer 6** | **MW UCAP** | - | - | 150 | 150 | 219 |
| | **$/MW-Day** | $0 | $0 | $300 | $300 | $300 |
| | **Revenue Cost Stream** | $0 | $0 | $45,000 | $45,000 | $65,575 |
| **Offer 7** | **MW UCAP** | - | - | - | 200 | 139 |
| | **$/MW-Day** | $0 | $0 | $0 | $300 | $300 |
| | **Revenue Cost Stream** | $0 | $0 | $0 | $60,000 | $41,734 |
| **Offer 8** | **MW UCAP** | - | - | - | 200 | 139 |
| | **$/MW-Day** | $0 | $0 | $0 | $300 | $300 |
| | **Revenue Cost Stream** | $0 | $0 | $0 | $60,000 | $41,734 |

The procurement process will prioritize earlier Delivery Years, while being constrained to the fixed-budget maximum willingness to pay. This can result in: (1) an earlier Delivery Year resource being skipped if it will violate the maximum willingness to pay constraint; (2) a procurement of that resource later in the process, even if the resource was previously skipped, if procurement of a lower-priced resource creates additional cost headroom in the budget; and/or (3) a possibility that PJM will not procure the exact Final RBP Target quantity if doing so violates the maximum willingness to pay constraint. These features are demonstrated below in the selection sequencing example:

Step 1: Procure Offer 1. This will add the below MW value to the target for each Delivery Year, with a MW-weighted average levelized costs of $100.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Total Offered** | 3,000 | 3,000 | 2,800 | 2,800 |

Step 2: Procure Offer 2. This will increase our procurement by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $93.26.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Total Offered** | 3,500 | 3,450 | 3,250 | 3,200 |

The MW-weighted average levelized cost of $93.26 can be calculated from the PV column in Table 2 as the sum of levelized MW / sum of Revenue cost stream. [9,322+1,453]/[932,200 + 72,645]

Step 3: Skip Offer 3. If Offer 3 was selected the MW-weighted average levelized cost would rise to $102.84, above the maximum willingness to pay of $100 MW/day.

Step 4: Procure Offer 4. This will increase the procured MW by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $80.50.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 |
| **Total Offered** | 3,500 | 5,450 | 5,150 | 5,200 |

Step 5: Procure Offer 3. Offer 3 may now be procured while keeping the MW-weighted average levelized cost under the maximum willingness to pay. Thus, the procurement will prioritize and commit this resource with an earlier first Delivery Year. This will increase the procured MW by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $88.27.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Offer 3** | | 400 | 500 | 500 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 |
| **Total Offered** | 3,500 | 5,850 | 5,650 | 5,700 |

Step 6: Procure Offer 5. This will increase the procured MW by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $88.47.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Offer 3** | | 400 | 500 | 500 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 |
| **Offer 5** | | 900 | 1,000 | 1,000 |
| **Total Offered** | 3,500 | 6,750 | 6,650 | 6,700 |

Step 7: Procure Offer 6. This will increase the procured MW by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $90.94

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Offer 3** | | 400 | 500 | 500 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 |
| **Offer 5** | | 900 | 1,000 | 1,000 |
| **Offer 6** | | | 150 | 150 |
| **Total Offe** | 3,500 | 6,750 | **6,800** | 7,250 |

At this point, the final target MW of 6,800 has been committed and the procurement is complete. Offer 7 and 8 would not be procured through the RBP.

## Example 3

In Example 3, the selection clears short of the Final RBP Target to maintain the maximum willingness to pay. This example relies on the same example as above, except that offer prices for Offers 6, 7, and 8 are dramatically increased.

*Table 3*

| | | 28/29 | 29/30 | 30/31 | 31/32 | PV |
|---|---|---|---|---|---|---|
| | **MW UCAP** | 3,000 | 3,000 | 2,800 | 2,800 | 9,322 |
| **Offer 1** | **$/MW-Day** | $100 | $100 | $100 | $100 | $100 |
| | **Revenue Cost Stream** | $300,000 | $300,000 | $280,000 | $280,000 | $932,200 |
| | **MW UCAP** | 500 | 450 | 450 | 400 | 1,453 |
| **Offer 2** | **$/MW-Day** | $50 | $50 | $50 | $50 | $50 |
| | **Revenue Cost Stream** | $25,000 | $22,500 | $22,500 | $20,000 | $72,645 |
| | **MW UCAP** | - | 400 | 500 | 500 | 1,062 |
| **Offer 3** | **$/MW-Day** | $0 | $200 | $200 | $200 | $200 |
| | **Revenue Cost Stream** | $0 | $80,000 | $100,000 | $100,000 | $212,444 |
| | **MW UCAP** | - | 2,000 | 1,900 | 2,000 | 4,506 |
| **Offer 4** | **$/MW-Day** | $0 | $50 | $50 | $50 | $50 |
| | **Revenue Cost Stream** | $0 | $100,000 | $95,000 | $100,000 | $225,316 |
| | **MW UCAP** | - | 900 | 1,000 | 1,000 | 2,208 |
| **Offer 5** | **$/MW-Day** | $0 | $90 | $90 | $90 | $90 |
| | **Revenue Cost Stream** | $0 | $81,000 | $90,000 | $90,000 | $198,705 |
| | **MW UCAP** | - | - | 150 | 150 | 219 |
| **Offer 6** | **$/MW-Day** | $0 | $0 | **$1,500** | **$1,500** | $1,500 |
| | **Revenue Cost Stream** | $0 | $0 | $225,000 | $225,000 | $327,876 |
| | **MW UCAP** | - | - | - | 200 | 139 |
| **Offer 7** | **$/MW-Day** | $0 | $0 | $0 | **$1,500** | $1,500 |
| | **Revenue Cost Stream** | $0 | $0 | $0 | $300,000 | $208,672 |
| | **MW UCAP** | - | - | - | 200 | 139 |
| **Offer 8** | **$/MW-Day** | $0 | $0 | $0 | **$2,000** | $2,000 |
| | **Revenue Cost Stream** | $0 | $0 | $0 | $400,000 | $278,230 |

The procurement process runs through Steps 1 through 5, as described in Example 2, above. Then, starting in Step 6, the process begins to differ:

Step 6: Procure Offer 5. This will increase the procured MW by the offered MW for each applicable Delivery Year, and result in a MW-weighted average levelized cost of $88.47.

| | 28/29 | 29/30 | 30/31 | 31/32 |
|---|---|---|---|---|
| **Offer 1** | 3,000 | 3,000 | 2,800 | 2,800 |
| **Offer 2** | 500 | 450 | 450 | 400 |
| **Offer 3** | | 400 | 500 | 500 |
| **Offer 4** | | 2,000 | 1,900 | 2,000 |
| **Offer 5** | | 900 | 1,000 | 1,000 |
| **Total Offered** | 3,500 | 6,750 | 6,650 | 6,700 |

At this point, Offers 6, 7 and 8 cannot be procured without violating the maximum willingness to pay constraint. If Offer 6 were selected, the MW-weighted average levelized cost would rise to $104.91, which is above the maximum willingness to pay of $100 MW/day. Similar violations of the maximum willingness to pay for Offers 7, and 8 would occur. Given that, in this example, there are no additional offers to consider, the procurement would clear 100 MW short the target, with the maximum UCAP MW procured equal to 6,700 MW for the 2031/32 Delivery Year.

**UNITED STATES OF AMERICA**
**BEFORE THE**
**FEDERAL ENERGY REGULATORY COMMISSION**

| | | |
|---|---|---|
| **PJM Interconnection, L.L.C.** | ) | **Docket No. ER26-____-000** |

## VERIFICATION

I, Danielle Croop, pursuant to 28 U.S.C. § 1746, state, under penalty of perjury, that I am the Danielle Croop referred to in the foregoing document entitled "Affidavit of Danielle Croop, on Behalf of PJM Interconnection, L.L.C.," that I have read the same and am familiar with the contents thereof, and that the facts set forth therein are true and correct to the best of my knowledge, information, and belief.

/s/ Danielle Croop
Danielle Croop
Sr. Manager
Market Design
PJM Interconnection, L.L.C.

Subscribed and sworn to before me, the undersigned notary public, this 31st day of July 2026.

Michele A. Matticola
Notary Public

My Commission expires: 12/23/2027

Commonwealth of Pennsylvania - Notary Seal
MICHELE A MATTICOLA - Notary Public
Montgomery County
My Commission Expires December 23, 2027
Commission Number 1249359